Project 2: Long/Short Factor-Based GTAA Portfolio

This project involves constructing and back-testing a **factor-based long-short global tactical asset allocation (GTAA) portfolio**.

Your team is required to submit the following by **May 8 at 11:59 PM** (just before midnight):

1. **Report**, documenting your inputs, approach and results (no more than 5 pages).
2. **Excel spreadsheet** containing key results and formulas, following the guidelines below.
3. **Programming code**, if applicable.

Project Requirements

Your portfolio must be based on **at least two factors**. Your backtest must cover a **minimum of 10 years**, with new **target portfolio weights derived at least monthly**.

The project consists of the following components:

1. Select investment universe

- Your **investment universe** can include the 10 developed country equity index futures**,** for which historical returns were provided.
- Alternatively, you may define your own investment universe and obtain relevant data.

2. Construct Factor-Mimicking Portfolios (FMPs)

- Construct FMPs for your chosen investment universe.
- You may use factors from the literature referenced in the lecture notes or create your own.
- Your portfolio must be based on **at least two factors**, with the following restriction:
    - Only **one factor can be either value or momentum**, as included in the provided examples and in the homework.

3. Estimate Covariance Matrices

- Estimate and update the **covariance matrix** for your assets at least every month.
- Each estimation must use at least **36 months of prior returns**.
- A **significant penalty** will be applied if you use too few data points, such as only the last 12 months of monthly returns to estimate the covariance matrix, which may lead to unreliable estimates.

4. Scale FMP Weights

- Scale weights for **each factor** to target **1% annual volatility**.
- **Combine all FMPs** using a **weighting scheme of your choice** (e.g., equal weighting).
- Rescale the **combined weights** again to target **1% annual volatility**.

5. Backtest Your Portfolio

Calculate monthly (or higher frequency depending on your data) returns on your portfolio and compute the following **summary statistic over the full backtest period (formulas must be included in your Excel spreadsheet)**:

- **Cumulative gross returns (labeled "Gross Returns")**:
  - Growth of $1 for your portfolio as well as for each FMP separately, also present as charts.
- **Portfolio Statistics (labeled "Portfolio Statistics"):**
  - Arithmetically annualized simple return
  - Annualized volatility of simple returns
  - Annualized information ratio
  - Average drawdown
  - Maximum drawdown
  - Turnover

6. Provide Additional Quantities (for the **entire period** of the backtest)

- **"Raw Factor Weights":** Raw asset weights for each of your FMPs before scaling to 1%, **over time**.
- **"Factor1 1% Volatility Weights" and "Factor2 1% Volatility Weights":** Asset weights for each of the two FMPs, scaled to 1% risk, **over time**, also presented as charts.
- **"Final 1% Volatility Portfolio Weights"**: Asset weights for your portfolio scaled to 1% risk over time, also presented as charts.

7. Provide Additional Quantities (for the **final time period** only**)**

- **"Covariances":** Annualized covariance matrix that you used to construct the final portfolio of your backtest.
- **"Volatilities":** Annualized volatility of each asset extracted from the above final covariance matrix.
- **"Correlations":** Correlation matrix, extracted from the above final covariance matrix.

Report Guidelines

Your **report** should describe your inputs, approach and results. It may include:

- **Data Sources:** Description of the data you used.
- **Investment Universe**: If you customized asset selection.
- **Factors**: Explanation of the selected factors.
- **Backtest Summary**: Key observations and insights from your backtest results.

Note 1: Your project grade is **not** based on portfolio performance but rather on the soundness of your assumptions, the rigor of your methodology, and the accuracy of your calculations.

Note 2: A **significant penalty** will be imposed for **look-ahead bias**. This includes using data that would not have been available at the time of portfolio construction, such as returns from the same month for which the portfolio is being built, as this compromises the integrity of the backtest.